HelixConstitution Gates — Lava Implementation Status

Last updated: 2026-05-16 (HelixQA Phase 4-debt closed — upstream PR `b13ba7c` landed `helix-deps.yaml` + `install_upstreams.sh`; Lava-side waivers removed; 17/17 own-org submodules satisfy `CM-HELIX-DEPS-MANIFEST` + `CM-CANONICAL-ROOT-CLARITY` + `CM-INSTALL-UPSTREAMS-RAN` in fully STRICT mode)
Inheritance: Per Lava `CLAUDE.md` §6.AD (HelixConstitution Inheritance Mandate); enumerated per §6.AD-debt item 8.

Purpose

HelixConstitution defines a set of `CM-*` mechanical gates (see `constitution/Constitution.md` §11.4 sub-clauses). Each gate has a status: ✅ wired, ⚠️ paper-only, or ❌ blocked. This document is the operator-readable index — the source of truth for "what's actually enforced versus what's documented."

Gate Inventory

Gate ID	HelixConstitution clause	Lava implementation	Status
`CM-COVENANT-114-16-PROPAGATION`	§11.4.16 (Item-Type tracking)	n/a (Lava uses `.lava-ci-evidence/crashlytics-resolved/` + `.lava-ci-evidence/sixth-law-incidents/` rather than Issues/Issues_Summary trackers; equivalence documented in §6.AD.3)	⚠️ pape only
`CM-COMMIT-DOCS-EXISTS`	§11.4.22 (lightweight commit path)	`scripts/commit_all.sh` exists + executable; verified by `scripts/check-constitution.sh` 6.AD(5)	✅ wire
`CM-FIXED-COLUMN-ALIGNMENT`	§11.4.19 (Fixed-document column alignment)	n/a (Lava does not maintain Fixed.md / Fixed_Summary.md tracker structure; functional equivalent: per-issue closure logs in `.lava-ci-evidence/crashlytics-resolved/`)	⚠️ pape only
`CM-SCRIPT-DOCS-SYNC`	§11.4.18 (script documentation mandate)	`.githooks/pre-push` Check 9 — fires when `scripts/X.sh` is modified without companion `docs/scripts/X.sh.md` in the same commit (only when the doc already exists at this commit); rehearsed in synthetic fixture	✅ wire
`CM-BUILD-RESOURCE-STATS-TRACKER`	§11.4.24 (build-resource stats tracking)	`scripts/build-stats-sample.sh` (per-sample collector, BSD+GNU portable, wraps a build via `wrap "<name>" -- <cmd>`) +	✅ wire

Gate ID	HelixConstitution clause	Lava implementation	Status
		`.lava-ci-evidence/build-stats/registry.tsv` (per-build summary registry) + `scripts/build-stats-report.sh` (derives `docs/build-stats/Stats.md` + `Stats.html` via pandoc + `Stats.pdf` via wkhtmltopdf-or-weasyprint; HTML/PDF skip silently if exporters absent)	
`CM-ITEM-STATUS-TRACKING`	§11.4.15 (item-status tracking)	n/a (Lava uses CONTINUATION.md + closure logs; mapping documented in §6.AD.3)	⚠️ pape only
`CM-ITEM-TYPE-TRACKING`	§11.4.16 (item-type tracking)	n/a (same as above)	⚠️ pape only
`CM-ITEM-OPERATOR-BLOCKED-DETAILS`	§11.4.21 (operator-blocked status + self-resolution audit)	n/a (Lava uses `.lava-ci-evidence/sixth-law-incidents/<date>.json` for blocker tracking; the `Operator-blocked` status vocabulary is not in use)	⚠️ pape only
`CM-OPERATOR-BLOCKED-SELF-RESOLUTION-AUDIT`	§11.4.21	n/a (same as above)	⚠️ pape only
`CM-UNIVERSAL-VS-PROJECT-CLASSIFICATION`	§11.4.17 (universal-vs-project classification)	`.githooks/pre-push` Check 8 — fires when `##### 6.X` clause added to CLAUDE.md without `Classification:` line in commit body	✅ wire
`CM-SUBAGENT-DELEGATION-AUDIT`	§11.4.20 (subagent-driven-by-default mandate)	n/a (no automated audit of subagent vs. foreground delegation in Lava; would require either Claude Code telemetry integration or manual operator review)	⚠️ pape only
`CM-LVA-TICKETS-SYNC`	§11.4.93 (SQLite single-source-of-truth) + §11.4.95 (DB tracked-in-git, never gitignored) + §11.4.106 (byte-identical doc⍰DB round-trip) + §11.4.33/34 (closure + reopen schema guards)	`scripts/check-lva-tickets.sh` (verifies (a) `docs/tickets/tickets.db` exists + is NOT gitignored, (b) the REAL `tools/lava-tickets verify` binary reports byte-identical trackers — §6.A real-binary contract, not a bash re-impl, (c) the five §11.4.33/34 schema-integrity triggers are present via a `sqlite_master` query; builds the binary via `go build` if absent; discovers `sqlite3` on PATH/Homebrew/Android platform-tools; `LAVA_LVA_TICKETS_STRICT=0` / `--advisory` for incremental adoption). Hermetic test: `tests/check-lva-tickets/test_check_lva_tickets.sh` (10 sub-tests: clean PASS + falsifiability — corrupt-tracker FAIL/restore PASS, drop-§11.4.33-trigger FAIL, drop-§11.4.34-trigger FAIL, gitignored-DB FAIL/un-ignore PASS, missing-DB FAIL, advisory exits 0). Evidence: `tests/check-lva-tickets/EVIDENCE.md`. Wired into `scripts/verify-all-constitution-rules.sh` in `--advisory` mode (build-	✅ wire (advisor

Gate ID	HelixConstitution clause	Lava implementation	Status
		advisory-then-strict-flip convention; strict-flip OWED after baseline stabilizes)	
`CM-UNIVERSAL-MARKDOWN-EXPORT-SYNC`	HelixConstitution §11.4.65 Universal Markdown Export Mandate	`scripts/sync-markdown-exports.sh` (`--check-only`/`--regenerate-all`/`--regenerate` <f>; pandoc HTML + `pandoc --pdf-engine=weasyprint` PDF; `timeout 60`/file; 500-candidate cap with log) + `scripts/check-markdown-export-sync.sh` (gate wrapper; `LAVA_MARKDOWN_EXPORT_STRICT` advisory/strict). Scope: project-root + `docs/**` + `scripts/**` `.md`; excludes source trees + submodules + constitution (they sync their own). Backfill 2026-05-31: 126 in-scope docs, generated `.html`+`.pdf` siblings (TRACKED per §11.4.65). Hermetic test: `tests/check-markdown-export-sync/test_markdown_export_sync.sh` (positive + delete-sibling-FAIL + backdate-mtime-FAIL + restore-PASS). Closes §6.AF-debt §11.4.65 item.	✅ wire
`CM-NONFATAL-COVERAGE`	§6.AC + HelixConstitution telemetry discipline	`scripts/check-non-fatal-coverage.sh` (bash scanner; default STRICT-mode after queue drained 2026-05-14; `LAVA_NONFATAL_STRICT=0` to revert to advisory mode)	✅ wire
`CM-CHALLENGE-DISCRIMINATION`	§6.AB Anti-Bluff Test-Suite Reinforcement	`scripts/check-challenge-discrimination.sh` (bash scanner; STRICT-default; verifies every Challenge*Test.kt carries a FALSIFIABILITY REHEARSAL marker in KDoc OR a companion .lava-ci-evidence/sp3a-challenges/.json; `LAVA_CHALLENGE_DISCRIMINATION_STRICT=0` to revert to advisory)	✅ wire
`CM-CHALLENGE-COVERAGE`	§6.AE Comprehensive Challenge Coverage + Container/QEMU Matrix Mandate	`scripts/check-challenge-coverage.sh` (bash scanner; STRICT-default after queue drained 2026-05-15; verifies every feature/* module has at least one Challenge that targets it via import / `// covers-feature:` marker / heuristic / `// AE-exempt:` ledger entry; `LAVA_CHALLENGE_COVERAGE_STRICT=0` to revert to advisory) + `scripts/run-challenge-matrix.sh` (matrix-runner glue; delegates to `submodules/containers/cmd/emulator-matrix --runner=containerized`; pre-bakes the §6.AE.2 minimum API/form-factor matrix; refuses to claim success on hosts lacking KVM with a clear §6.X-debt host-gap diagnostic). Hermetic test: `tests/check-constitution/`	✅ wire (PARTI scanner STRICT and-gre runner s BLOCK on darw arm64 p standing §6.X-de

Gate ID	HelixConstitution clause	Lava implementation	Status
		`test_challenge_coverage.sh` (4 falsifiability fixtures, all PASS)	
`CM-VERIFY-ALL-CONSTITUTION-RULES`	HelixConstitution §11.4.32 Post-Constitution-Pull Validation Mandate	`scripts/verify-all-constitution-rules.sh` (master sweep; STRICT default; wraps every individual gate above + every hermetic test suite into a single enforcement engine; emits per-run attestation JSON at `.lava-ci-evidence/verify-all/<UTC-timestamp>.json` with per-gate pass/fail/duration); wired into `scripts/ci.sh` step 5a4. Hermetic test: `tests/check-constitution/test_verify_all_rules.sh` (4 falsifiability fixtures including the §11.4.32-mandated paired-mutation that plants a known violation + asserts sweep reports FAIL). PER §11.4.32 ITSELF: this gate is "the enforcement engine for every other §11.4.x and CONST-NNN rule" — without it, new rules cascade as anchors but never get enforced in the codebase	✅ wire (Phase constitu complia plan, co `4def2d`
`CM-GITIGNORE-PRECOMMIT-AUDIT`	HelixConstitution §11.4.30 .gitignore + No-Versioned-Build-Artifacts Mandate	`scripts/check-gitignore-coverage.sh` (STRICT default; verifies every actual Gradle module + every owned-by-us submodule has .gitignore; verifies no tracked files match the forbidden-pattern set with explicit allowlist for documented-non-sensitive deployment configs); wrapped by verify-all sweep. Hermetic test: `tests/check-constitution/test_gitignore_coverage.sh` (3 fixtures: real-tree-passes, module-without-gitignore-rejected, clean-fixture-passes)	✅ wire (Phase 2 constitu complia plan)
`CM-NO-NESTED-OWN-ORG-SUBMODULES`	HelixConstitution §11.4.28 Submodules-As-Equal-Codebase Mandate	`scripts/check-no-nested-own-org-submodules.sh` (STRICT default; walks every owned submodule's `.gitmodules`; rejects entries pointing to vasic-digital / HelixDevelopment / red-elf / ATMOSphere1234321 / Bear-Suite / BoatOS123456 / Helix-Flow / Helix-Track / Server-Factory orgs on github.com or gitlab.com; per-entry waiver mechanism with mandatory rationale + tracking-issue + target-removal-date); wrapped by verify-all sweep in STRICT mode. Hermetic test: `tests/check-constitution/test_nested_own_org_submodules.sh` (4 fixtures: clean-fixture-passes, vasic-digital-chain-rejected, HelixDevelopment-gitlab-chain-rejected, advisory-mode-returns-zero). Phase 5-debt closed 2026-05-15: github cascade merge removed Challenges/.gitmodules (CONST-051(C) flat-layout enforcement);	✅ wire STRIC (Phase constitu complia plan)

Gate ID	HelixConstitution clause	Lava implementation	Status
		Panoptic no longer nested; scanner reports 0 violations in STRICT mode	
`CM-CANONICAL-ROOT-CLARITY` + `CM-INSTALL-UPSTREAMS-RAN`	HelixConstitution §11.4.35 Canonical-Root Inheritance Clarity + §11.4.36 Mandatory install_upstreams	`scripts/check-canonical-root-and-upstreams.sh` (STRICT default; verifies (a) root CLAUDE.md + AGENTS.md open with `## INHERITED FROM constitution/<file>` or `@constitution/<file>` pointer in first 40 lines, (b) constitution submodule's three canonical files exist, (c) constitution submodule's own CLAUDE.md/AGENTS.md do NOT carry `## INHERITED FROM` at top level — fenced ` ```markdown ` code blocks correctly recognized as documentation; verifies every owned submodule has install_upstreams script at one of 6 supported locations; per-submodule waiver mechanism with mandatory rationale); wrapped by verify-all sweep in STRICT mode. Hermetic test: `tests/check-constitution/test_canonical_root_and_upstreams.sh` (6 fixtures including discrimination test for fenced-code-block false-positive prevention). Phase 8-debt closed 2026-05-15: 10 install_upstreams scripts landed (Challenges, Config, Containers, Discovery, HTTP3, Mdns, Middleware, RateLimiter, Recovery, Tracker-SDK each gained install_upstreams.sh + Upstreams/GitHub.sh + Upstreams/GitLab.sh per §6.W). Phase 4-debt closed 2026-05-16: HelixQA upstream PR `b13ba7c` (`HelixDevelopment/HelixQA`) landed `install_upstreams.sh` wrapper at HelixQA's repo root + Lava parent removed HelixQA from `INSTALL_UPSTREAMS_WAIVERS`; scanner now reports 17/17 install_upstreams present (0 waived) + all §11.4.35 sub-checks passing in STRICT mode	✅ wire STRICT (Phase 8 constitu complia plan; Ph 4-debt c 2026-05
`CM-HELIX-DEPS-MANIFEST`	HelixConstitution §11.4.31 Submodule-Dependency-Manifest Mandate	`scripts/check-helix-deps-manifest.sh` (STRICT default; verifies (a) parent project root has `helix-deps.{yaml,json,toml}` with `schema_version: 1` + `deps:` + `transitive_handling:` blocks, (b) every owned-by-us submodule has its own `helix-deps.{yaml,json,toml}`; per-submodule waiver mechanism with mandatory rationale); wrapped by verify-all sweep in STRICT mode. Hermetic test: `tests/check-constitution/test_helix_deps_manifest.sh` (6 fixtures: compliant-fixture-passes, missing-parent-rejected, missing-submodule-rejected, wrong-schema-version-rejected, advisory-mode-returns-zero, json-variant-accepted). Phase 3-	✅ wire STRICT (Phase 1 constitu complia plan; Ph 4-debt c 2026-05

Gate ID	HelixConstitution clause	Lava implementation	Status
		debt closed 2026-05-15: 16/16 per-submodule helix-deps.yaml manifests landed (Auth, Cache, Challenges, Concurrency, Config, Containers, Database, Discovery, HTTP3, Mdns, Middleware, Observability, RateLimiter, Recovery, Security, Tracker-SDK — each authored with honest per-submodule deps after analysis: 15 leaf modules with empty deps + Challenges declaring Panoptic with layout: flat). Phase 4-debt closed 2026-05-16: HelixQA upstream PR `b13ba7c` (`HelixDevelopment/HelixQA`) landed `helix-deps.yaml` at HelixQA's repo root (declares Challenges + Containers deps per HelixQA's README sibling-directory enumeration) + Lava parent removed HelixQA from `HELIX_DEPS_WAIVERS`; parent helix-deps.yaml declares 18 deps (16 vasic-digital + HelixDevelopment/HelixQA + constitution); scanner now reports 17/17 per-submodule manifests present (0 waived) in STRICT mode	
`CM-COVERAGE-LEDGER`	HelixConstitution §11.4.25 Full-Automation-Coverage Mandate	`scripts/generate-coverage-ledger.sh` (deterministic per-row YAML generator; walks `feature/*/`, `core/*/`, `app/`, `lava-api-go/`, `submodules/*/`; emits 6 invariants per row — anti_bluff, working_capability, documented_promise, no_open_bugs, documented, four_layer — with status `pass`/`gap`/`n/a`/`waiver:`; produces `docs/coverage-ledger.yaml`) + `scripts/check-coverage-ledger.sh` (verifier; asserts schema-presence, every-on-disk-path-has-a-row, AND ledger-fresh-vs-regenerated). Companion waiver mechanism: `docs/coverage-ledger.waivers.yaml` (hand-edited per-row overrides for invariants 3/4 the generator cannot mechanically infer). Hermetic test: `tests/check-constitution/test_coverage_ledger.sh` (6 fixtures: clean-fixture-passes, missing-ledger-rejected, missing-row-rejected, stale-ledger-rejected, advisory-mode-returns-zero, generator-deterministic). Default: --advisory in verify-all sweep until per-row backfill stabilises baseline; STRICT flip OWED via Phase 7-debt	✅ wire (advisor (Phase constitu complia plan)
`CM-CLOSURE-STATUS-VOCAB-COMPLIANCE`	HelixConstitution §11.4.33 Type-aware closure-status vocabulary	EQUIVALENCE-MAPPED per §6.AD.3 (Phase 9 Path B): Lava's `.lava-ci-evidence/sixth-law-incidents/<date>-<slug>.json` schema includes candidate-causes, ELIMINATED entries, and `PENDING_FORENSICS:` markers (per §11.4.6)	✅ equivale mapped (Phase 9 Path B constitu

Gate ID	HelixConstitution clause	Lava implementation	Status
		that satisfy the type-aware closure-status semantics §11.4.33 prescribes. No standalone scanner — the §6.AD.3 mapping IS the binding equivalence per Path B.	complia plan)
`CM-REOPENED-SOURCE-ATTRIBUTION`	HelixConstitution §11.4.34 Reopened-source attribution	EQUIVALENCE-MAPPED per §6.AD.3 (Phase 9 Path B): Lava's `.lava-ci-evidence/crashlytics-resolved/<date>-<slug>.md` closure-log schema includes root-cause + fix-commit-SHA + validation-test references — providing WHY + WHO + WHEN + WHICH-INCIDENT lineage for each closed item. Re-opens are tracked as new closure-log entries citing the prior log. No standalone scanner — the §6.AD.3 mapping IS the binding equivalence per Path B.	✅ equival mapped (Phase 9 Path B c constitu complia plan)
`CM-HELIXQA-WIRING`	HelixConstitution §11.4.27 (CONST-050) full test-type coverage including HelixQA + Lava §6.AE Comprehensive Challenge Coverage	`scripts/run-helixqa-challenges.sh` (Lava-side wrapper invoking the 11 HelixQA Challenge scripts at `submodules/helixqa/challenges/scripts/`; captures per-script stdout+stderr to log files; emits roll-up attestation JSON; classifies exit codes 0/2/other → PASS/SKIP/FAIL; refuses to silently mask missing submodule or wiring drift — exit 3 with actionable remediation per §6.J anti-bluff posture). Wired into `scripts/run-challenge-matrix.sh` via `--include-helixqa` opt-in flag (OFF by default so existing matrix runs are unaffected). Phase 4 follow-up B (2026-05-16) added: `--runner=host|containerized` flag (§6.X container delegation, honest-fail-fast exit 4 on missing Containers submodule), `--require-toolchain=go|none` flag (§6.J toolchain precondition gate with `HELIXQA_TOOLCHAIN_MAP` per-script declaration), `LAVA_HELIXQA_EVIDENCE_DIR` env-var override (Q3), `HELIXQA_W_EXCLUSIONS` array consuming `docs/helixqa-script-audit.md` (Q4). Hermetic test: `tests/check-constitution/test_helixqa_wiring.sh` (11 fixtures covering all 4 questions' resolutions + the original 4 fixtures). Docs: `docs/scripts/run-helixqa-challenges.sh.md` + `docs/scripts/run-challenge-matrix.sh.md` + `docs/helixqa-script-audit.md`. Design: `docs/plans/2026-05-16-helixqa-integration-design.md` Option 1 — shell-level wiring. Many HelixQA scripts are Linux-only (systemctl/journalctl); macOS hosts will see FAIL/SKIP outcomes that are HONEST	✅ wire (Option shell wi + 4 ope question resoluti host run is defau contain runner r for §6.A gate run

Gate ID	HelixConstitution clause	Lava implementation	Status
		acknowledgments of the wrong-OS gap rather than false-pass.	
`CM-HELIXQA-§6.W-AUDIT`	Lava §6.W (GitHub + GitLab Only Remote Mandate) + §6.J (Anti-Bluff Functional Reality)	`docs/helixqa-script-audit.md` (per-script audit of all 11 HelixQA Challenge scripts; 4-query methodology: git push / curl-wget / external HTTP / outside-worktree FS writes; per-script verdict table with COMPLIANT / REQUIRES-EXCLUSION / HOST-MUTATION-RISK classification; aggregate verdict + operator-supplied env-var risk surface + re-audit triggers). Consumed by `scripts/run-helixqa-challenges.sh` via the `HELIXQA_W_EXCLUSIONS` array constant (currently empty since 0/11 scripts violate §6.W on default config). Re-audit OWED whenever the submodules/helixqa pin bumps; the wrapper's wiring-drift check (exit 3 on added/removed scripts) prompts re-audit but does not mechanize the per-script grep — that remains a human-driven audit per §6.J ("real grep results only — no manufactured findings").	✅ wire (advisor per-scri audit is driven; mechan scanner OWED future p bump introduc violator

Lava-side anti-bluff gates (NOT HelixConstitution `CM-*` but related)

For completeness — these gates predate or extend HelixConstitution's `CM-*` set:

Gate name	Lava clause	Implementation	Status
Hosted-CI config rejection	Local-Only CI/CD rule	`.githooks/pre-push` Check 1	✅ wired
Test-commit Bluff-Audit stamp	Seventh Law clause 1	`.githooks/pre-push` Check 2	✅ wired
Mock-the-SUT pattern rejection	Seventh Law clause 4	`.githooks/pre-push` Check 3	✅ wired
Gate-shaping file Bluff-Audit stamp + path-match	§6.N.1.2	`.githooks/pre-push` Check 4 (with extended `.githooks/pre-push` regex)	✅ wired
Attestation falsifiability rehearsal	§6.N.1.3	`.githooks/pre-push` Check 5	✅ wired
§6.Y bump-first ordering	§6.Y	`.githooks/pre-push` Check 6	✅ wired
§6.Z evidence-file presence on pointer advance	§6.Z	`.githooks/pre-push` Check 7 + `firebase-`	✅ wired

Gate name	Lava clause	Implementation	Status
		`distribute.sh` Phase-1 Gates	
§6.AA Two-Stage default + release-requires-debug-companion	§6.AA	`firebase-distribute.sh` (default flipped to `MODE=debug`; `--release-only` requires `last-version-debug >= current vc`)	✅ wired
§6.W mirror-host boundary	§6.W + §6.AD.1	`scripts/check-constitution.sh` 6.AD §6.W block	✅ wired
§6.AD per-scope inheritance pointer-block presence	§6.AD.4	`scripts/check-constitution.sh` 6.AD(4); 54 in-scope files enforced	✅ wired
§6.AD constitution submodule presence + structure	§6.AD.2	`scripts/check-constitution.sh` 6.AD(2)	✅ wired
§6.AD `commit_all.sh` + `inject-helix-inheritance-block.sh` presence	§6.AD.5 + §6.AD.6	`scripts/check-constitution.sh` 6.AD(5)+(6)	✅ wired
§11.4.6 no-guessing vocabulary	§6.AD.6 + HelixConstitution §11.4.6	`scripts/check-constitution.sh` final block	✅ wired
§6.X Container-Submodule Emulator wiring	§6.X	`scripts/check-constitution.sh` 6.X(4)+(5) (with PARTIAL CLOSE — Linux x86_64 gate-host still owed)	✅ wired (PARTIAL)
§6.K Containers extension presence	§6.K	`scripts/check-constitution.sh` clause 6.K block	✅ wired

Why some gates are ⚠️ paper-only

The `CM-*` gates marked ⚠️ paper-only typically depend on infrastructure Lava does not have today:

- **Issues / Issues_Summary / Fixed / Fixed_Summary trackers** (§11.4.15 / §11.4.16 / §11.4.19 / §11.4.21): HelixConstitution prescribes a project-wide work-tracker file structure. Lava's equivalent is `docs/CONTINUATION.md` (single-file source-of-truth handoff per §6.S) plus per-issue closure logs in

`.lava-ci-evidence/crashlytics-resolved/<date>-<slug>.md` (per §6.O). The substance is equivalent (status + type + history); the file structure differs. Lava does not plan to migrate to the HelixConstitution file structure unless / until the operator commissions that work; the equivalence is documented in §6.AD.3.

- **Build-resource stats tracker** (§11.4.24 / `CM-BUILD-RESOURCE-STATS-TRACKER`): Requires a host-side sampler that captures memory / CPU / IO at fixed interval, computes per-metric min/max/mean/p95, appends to a TSV registry, derives Stats.{md,html,pdf} via the §11.4.22 lightweight commit path. Substantial implementation work (~200+ lines plus a viewer). Tracked under §6.AD-debt item 5 for a future cycle.

- **Subagent-delegation audit** (§11.4.20 / `CM-SUBAGENT-DELEGATION-AUDIT`): Requires either Claude Code telemetry integration (not yet exposed by the platform) or operator-driven self-audit. Tracked under §6.AD-debt for a future cycle when telemetry surfaces become available.

- **Script documentation sync** (§11.4.18 / `CM-SCRIPT-DOCS-SYNC`): The doc files exist (`docs/scripts/commit_all.sh.md`, `docs/scripts/inject-helix-inheritance-block.sh.md`) but no pre-push gate yet enforces "modifying `scripts/X.sh` requires modifying `docs/scripts/X.sh.md` in the same commit". The pre-push gate is straightforward to add when the doc-set is more complete. Tracked under §6.AD-debt.

How to read this document

- ✅ wired: a script enforces the rule; pre-push or similar gate fires on violation; can demonstrate by running the falsifiability rehearsal cited in the original closure commit
- ⚠️ paper-only: the rule is documented in CLAUDE.md / AGENTS.md / Constitution.md but no mechanical gate enforces it; operator + reviewer manually verify
- ❌ blocked: the rule cannot be enforced today due to missing infrastructure; closure work is tracked under §6.AD-debt

How to wire a new gate

1. Read the corresponding `Constitution.md` §11.4.X clause for the rule's intent
2. Pick the enforcement surface: pre-push hook (`.githooks/pre-push`), constitution check (`scripts/check-constitution.sh`), distribute script (`scripts/firebase-distribute.sh`), tag script (`scripts/tag.sh`)
3. Add the gate logic; include `|| true` on every `grep -oE` pipe to avoid `set -e` killing the gate on zero matches
4. Add a falsifiability rehearsal: deliberately break a target file, re-run the gate, confirm the violation message includes the target path
5. Add a hermetic test under `tests/<area>/` exercising both the positive and negative paths
6. Update this document — flip ⚠️ to ✅ for the gate's row
7. Commit + push via `scripts/commit_all.sh` with proper Bluff-Audit stamp + Classification line

Cross-references

- Lava `CLAUDE.md` §6.AD (HelixConstitution Inheritance Mandate) + §6.AD-debt
- HelixConstitution `Constitution.md` §11.4.1 through §11.4.24
- HelixConstitution `CLAUDE.md` (universal CLAUDE.md inherited by every consuming project)